

# Expiry-Aware Successor Selection

Technical Methodology and Results

BOAMP Grand Ouest Digital Procurement Pipeline

BOAMP Data Science Internship Project

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## Abstract

This report documents the implemented expiry-aware successor-selection policy for the BOAMP Grand Ouest digital procurement study. The policy leaves broad candidate generation unchanged and modifies only candidate selection. A candidate published more than one year before the best observable expected contract end must satisfy stronger text and CPV continuity requirements. Applied to 3,800 cohort contracts and 868,527 candidate pairs, the policy accepts 507 observable successor procurements, compared with 547 under the frozen text-only baseline. Reviewed-benchmark metrics are unchanged: precision 0.8000, recall 0.3636, and zero false positives on reviewed negative anchors. The automatic promotion gate passes, but the method remains a parallel sensitivity arm until 42 changed links receive manual review. Accepted links are observable successor procurements, not confirmed legal renewals.

The underlying modeling phase is implemented, not proposed. Four pre-existing linkage methods were fitted or executed and compared before the expiry-aware fifth policy was added. This report includes their algorithms, operating points, benchmark results, and failure analysis so that implementation status can be audited directly.

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# 1 Technical Summary

The expiry-aware policy is more conservative and more explicitly aligned with contract timing than the frozen baseline because it addresses a documented failure mode: a public buyer can publish many unrelated procurements while an existing contract is active. The baseline `M_B_text_ranking` method accepts a same-buyer candidate primarily from text similarity, so repeated buyer and administrative language can produce an implausibly early link. The new policy retains broad retrieval but raises the evidence standard for candidates more than 365 days before expected expiry.

The implemented result is conservative:

- candidate generation remains fixed at 868,527 pairs;
- accepted links fall from 547 to 507;
- 505 baseline pairs are retained, 42 are removed, and 2 replacement pairs are selected;
- the survival event rate falls from 14.39% to 13.34%;
- median observed time to successor rises from 32.07 to 34.40 months;
- all four digital segments retain more than 30 observed events;
- benchmark precision, recall, and false-positive rate are unchanged.

The new method passes the pre-specified numerical promotion gate. This is not sufficient for automatic promotion: expiry is unavailable for 2,845 of 3,800 contracts, and the small benchmark does not independently validate most changed links. The operational decision is to keep the method parallel until the complete 42-row review queue is adjudicated.

## 2 Implemented Boundary and Data Scope

### 2.1 The change occurs after candidate generation

Figure 1 locates the new step within the implemented pipeline. Raw acquisition, standardisation, episode reconstruction, cohort definition, same-buyer blocking, and the broad temporal window are unchanged. The policy enriches existing candidate pairs with expiry evidence and applies an additional acceptance rule.

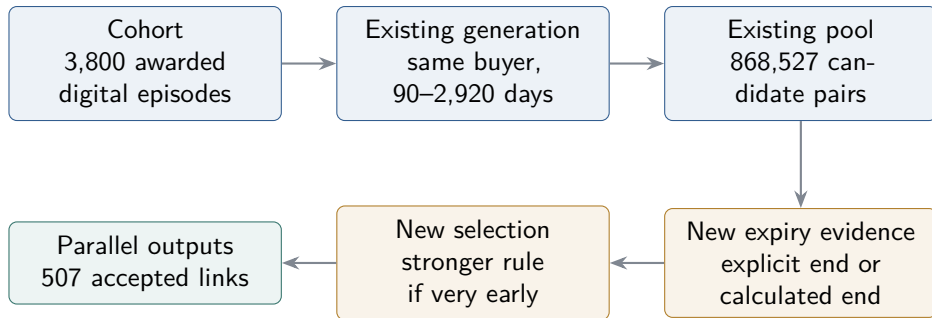


Figure 1: Pipeline boundary of the expiry-aware method. Blue stages are unchanged; gold stages are new; the green stage contains parallel v2 outputs.

### 2.2 Population, grain, and observation period

The study cohort contains 3,800 procurement episodes. Each episode is a Grand Ouest public procurement with at least one digital CPV division in  $\{32, 35, 48, 72\}$  and an award notice. One cohort row represents one awarded procurement episode. The award-notice publication date is time zero. The corpus and right-censoring cutoff end on 31 December 2025.

The candidate pool contains one row per ordered anchor–candidate episode pair. A candidate must belong to the same buyer block and be published between 90 and 2,920 days after the anchor award. Candidate episodes are not restricted to digital CPVs or award notices because a successor can first appear as a tender and may be incompletely coded.

### 3 Notation and Metric Definitions

Table 1 defines each symbol before it is used in the model specification.

Table 1: Notation for expiry-aware successor selection.

| Symbol      | Definition                                                                     |
|-------------|--------------------------------------------------------------------------------|
| $a$         | Anchor contract in the 3,800-row survival cohort.                              |
| $c$         | Later candidate procurement episode from the same buyer block.                 |
| $A_a$       | Award-notice publication date of anchor $a$ .                                  |
| $C_c$       | First-notice publication date of candidate $c$ .                               |
| $D_a$       | Reliable declared duration of anchor $a$ , measured in months.                 |
| $S_a$       | Unambiguous explicit contract start date, when observable.                     |
| $F_a$       | Unambiguous explicit contract end date, when valid and observable.             |
| $E_a$       | Best available expected end date of anchor $a$ .                               |
| $R_{ac}$    | Candidate timing relative to expected end, $C_c - E_a$ , in days.              |
| $s_t(a, c)$ | Maximum word- or character-TF-IDF cosine similarity in $[0, 1]$ .              |
| $s_k(a, c)$ | CPV continuity score; positive values indicate shared CPV hierarchy.           |
| $S(a, c)$   | Existing weighted linkage score in $[0, 100]$ , used as a tie-breaker.         |
| $m_j(\ell)$ | Probability of comparison level $\ell$ for field $j$ among matches.            |
| $u_j(\ell)$ | Probability of comparison level $\ell$ for field $j$ among non-matches.        |
| $\lambda$   | Fellegi–Sunter prior probability that a candidate pair is a match.             |
| $M(a, c)$   | Fellegi–Sunter $\log_2$ match weight for pair $(a, c)$ .                       |
| $\hat{y}_a$ | Selected observable successor for $a$ , or $\emptyset$ if the method abstains. |
| $\delta_a$  | Survival event indicator: 1 if a successor is selected, otherwise 0.           |
| $T_a$       | Observed time from award to successor, or award to cutoff when censored.       |

Let TP be a selected candidate that equals a reviewed successor, FP a selected wrong candidate or a selection on a reviewed no-successor anchor, and FN a reviewed successor not selected at rank one. Then

$$\text{Precision@1} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall@1} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (1)$$

The reported false-positive rate is defined on reviewed negative anchors:

$$\text{FPR}_- = \frac{\text{negative anchors receiving a link}}{\text{reviewed negative anchors}}. \quad (2)$$

## 4 Candidate Evidence and Implemented Algorithms

### 4.1 The model phase is materialised and executable

Table 2 separates implemented code and outputs from future work. Every row shown as implemented has an executable script, a materialised artifact, and automated validation or benchmark evidence.

Table 2: Implementation audit of the linkage-model phase.

| Component                   | State       | Material evidence                                                                            |
|-----------------------------|-------------|----------------------------------------------------------------------------------------------|
| Candidate features          | Implemented | 868,527 rows in <code>linkage_candidates.parquet</code> .                                    |
| Deterministic method $M_A$  | Implemented | Executed by <code>scripts/evaluate_linkage.py</code> .                                       |
| Text-ranking method $M_B$   | Implemented | Frozen v1 operating point at 0.70.                                                           |
| Weighted gated method $M_C$ | Implemented | Executed at score threshold 70 and retained as a high-recall sensitivity arm.                |
| Fellegi–Sunter $M_D$        | Fitted      | EM model in <code>fellegi_sunter_model.json</code> ; scored values for every candidate pair. |
| Expiry-aware method $M_E$   | Implemented | 507 accepted links plus the 42-anchor review artifact.                                       |
| Threshold evaluation        | Implemented | Pilot grid and all-method comparison CSVs.                                                   |
| Survival conversion         | Implemented | Parallel 3,800-row dataset with structural validation.                                       |

## 4.2 Four interpretable evidence components are computed

All methods operate on the same candidate pairs, but they use the evidence components differently.

**Buyer evidence.** Buyer matching distinguishes a validated SIREN match, an exact normalised-name match, fuzzy-name evidence, token overlap, and no match. The score mapping is

$$s_b(a, c) \in \{1.00, 0.90, 0.70, 0.60, 0.00\}, \quad (3)$$

in that order. Candidate blocking already requires the same exact buyer key or normalised blocking name; this score measures the strength of that identity evidence. Simple raw case variants such as `Ville de Paris` and `VILLE DE PARIS` normalise to the same name. Organisational suffixes remain distinct unless a shared validated SIREN or the blocking-name rule provides continuity; this is formatting normalisation, not complete buyer entity resolution.

**Text evidence.** Text is normalised to lowercase accent-free alphanumeric tokens. For each anchor block, separate word TF–IDF vectors with 1–2 grams and character TF–IDF vectors with 3–5 character grams are fitted. Let  $\mathbf{x}^{(w)}$  and  $\mathbf{x}^{(q)}$  denote word and character vectors. The implemented text score is

$$s_t(a, c) = \max \left\{ \frac{\mathbf{x}_a^{(w)} \cdot \mathbf{x}_c^{(w)}}{\|\mathbf{x}_a^{(w)}\| \|\mathbf{x}_c^{(w)}\|}, \frac{\mathbf{x}_a^{(q)} \cdot \mathbf{x}_c^{(q)}}{\|\mathbf{x}_a^{(q)}\| \|\mathbf{x}_c^{(q)}\|} \right\}. \quad (4)$$

Local fitting makes common boilerplate within one buyer less influential than rare terms describing the repeated purchasing need. Character grams preserve evidence when spelling, accents, abbreviations, or word boundaries vary.

**CPV continuity.** Let  $G_3(e)$  be the set of three-digit CPV groups in episode  $e$ ,  $G_2(e)$  its two-digit divisions, and  $J(X, Y) = |X \cap Y| / |X \cup Y|$  the Jaccard coefficient. Then

$$s_k(a, c) = \max \{J(G_3(a), G_3(c)), 0.5J(G_2(a), G_2(c))\}. \quad (5)$$

The score is missing, not zero, if either episode has no CPV. A zero means that both sides are coded and disagree; missing means continuity is unobservable.

**Timing plausibility.** Let  $g_{ac} = C_c - A_a$  be the award-to-candidate gap and let  $d_a = 30.4375D_a$  be expected duration in days. Where duration is reliable,

$$s_\tau(a, c) = \max \left\{ 0, 1 - \frac{|g_{ac} - d_a|}{365} \right\}. \quad (6)$$

The score is missing when duration is unresolved or conflicting. The implementation does not substitute a four-year default.

**Weighted score.** Let  $\mathcal{O}(a, c)$  be the observable component set. The implemented weights are shown in buyer, text, CPV, and timing order:

$$(w_b, w_t, w_k, w_\tau) = (0.50, 0.25, 0.20, 0.05). \quad (7)$$

The combined score is

$$S(a, c) = 100 \frac{\sum_{j \in \mathcal{O}(a, c)} w_j s_j(a, c)}{\sum_{j \in \mathcal{O}(a, c)} w_j}. \quad (8)$$

Renormalisation prevents missing evidence from acting like disagreement. It can also make a score rely on fewer dimensions, which is why  $M_C$  adds independent gates.

### 4.3 Five linkage methods have defined algorithms

**$M_A$ : deterministic SIREN-CPV rule.** The candidate is eligible only when the buyer has the same validated SIREN,  $s_k(a, c) > 0$ , and  $s_t(a, c) \geq 0.25$ . Eligible candidates are ranked by the weighted score. This tests what rigid structured identity and category continuity can achieve.

**$M_B$ : same-buyer text ranking.** Within the already blocked same-buyer pool,

$$I_B(a, c) = \mathbf{1}[s_t(a, c) \geq \theta_B]. \quad (9)$$

The frozen operating point is  $\theta_B = 0.70$ , and eligible candidates are ranked by  $s_t$ . This is the v1 baseline because it achieved the strongest reviewed precision and zero false positives on reviewed negative anchors.

**$M_C$ : weighted score with independent gates.** At the evaluated operating point,  $M_C$  requires all of the following:

- strong buyer evidence: SIREN, exact normalised name, or fuzzy-name similarity at least 0.82;
- $s_t \geq 0.25$ ;
- positive CPV continuity, or  $s_t \geq 0.55$  as a text override;
- if buyer evidence is weaker than SIREN or exact name,  $s_t \geq 0.40$  and positive CPV continuity;
- weighted score  $S(a, c) \geq 70$ .

Eligible candidates are ranked by  $S(a, c)$ . The method is designed to recover more successors while requiring evidence from several dimensions.

**$M_D$ : unsupervised Fellegi–Sunter model.** Buyer, text, CPV, and timing comparisons are discretised into levels. For field  $j$  and observed level  $\ell_j$ , the model estimates

$$m_j(\ell_j) = P(\ell_j \mid \text{match}), \quad u_j(\ell_j) = P(\ell_j \mid \text{non-match}). \quad (10)$$

Assuming conditional independence, evidence is accumulated as a log Bayes factor:

$$M(a, c) = \log_2 \frac{\lambda}{1 - \lambda} + \sum_j \log_2 \frac{m_j(\ell_j)}{u_j(\ell_j)}, \quad (11)$$

with posterior

$$P(\text{match} \mid a, c) = \frac{2^{M(a, c)}}{1 + 2^{M(a, c)}}. \quad (12)$$

Expectation maximisation alternates between pair-level match responsibilities and updated  $m$  distributions and  $\lambda$ . The primary fit fixes  $u$  to population comparison frequencies, uses Laplace smoothing 0.5, converges monotonically in 98 iterations, and scores every candidate pair. A free- $u$  fit is retained as a robustness check. Candidates are accepted by a posterior threshold and ranked by the posterior; the comparison point is 0.65.

**$M_E$ : expiry-aware text ranking.** The fifth method is the v2 policy specified in Section 6. It retains the  $M_B$  rule except that a candidate more than 365 days before expected end requires  $s_t \geq 0.85$  and  $s_k > 0$ . It is a fixed rule, not a retrained model.

## 5 Model Evaluation and Selection

### 5.1 The benchmark and retrieval stage are both evaluated

The reference data were remapped to v2 episode IDs through stable BOAMP notice IDs. Of 92 benchmark anchors, 85 are evaluable in the cohort: 22 positives and 63 reviewed no-successor anchors. The pilot-development split has 16 anchors, including only 5 positives; the already-inspected locked split has 69 anchors, including 17 positives.

Before method-specific acceptance, candidate retrieval places the true successor at rank one for 72.73% of positive anchors, within rank five for 86.36%, within rank ten for 86.36%, and within rank twenty for 90.91%. Twenty-one of 22 reviewed successors occur somewhere in the pool; the worst retrieved rank is 27. Thus the acceptance methods cannot recover one positive that candidate generation does not retrieve.

### 5.2 Thresholds were evaluated, not supervised-fitted

$M_A$ ,  $M_B$ , and  $M_C$  were evaluated over thresholds  $\{50, 55, 60, 65, 70, 75, 80\}$ .  $M_D$  used posterior thresholds  $\{20, 30, 40, 50, 55, 60, 65, 70\}$ . These are operating-point comparisons, not supervised model fine-tuning. With only five pilot positives, one changed prediction moves pilot recall by 0.20, so the frozen choice was documented judgement rather than an automatic maximum over noisy pilot metrics.

### 5.3 All five methods are compared at their implemented operating points

Table 3: All-method benchmark comparison on 85 evaluable anchors.

| Method               | Threshold   | Precision     | Recall@1      | FPR <sub>-</sub> | Coverage |
|----------------------|-------------|---------------|---------------|------------------|----------|
| $M_A$ deterministic  | rule        | 0.5556        | 0.4545        | 0.1270           | 0.2118   |
| $M_B$ text ranking   | 0.70        | <b>0.8000</b> | 0.3636        | <b>0.0000</b>    | 0.1176   |
| $M_C$ weighted gated | 70          | 0.5714        | <b>0.7273</b> | 0.1587           | 0.3294   |
| $M_D$ Fellegi–Sunter | 0.65        | 0.3750        | 0.2727        | 0.0952           | 0.1882   |
| $M_E$ expiry-aware   | 0.70 / 0.85 | <b>0.8000</b> | 0.3636        | <b>0.0000</b>    | 0.1176   |

Figure 2 makes the precision–recall trade-off visible.  $M_B$  and  $M_E$  prioritise precision and abstention;  $M_C$  prioritises recall at the cost of substantially more false positives.

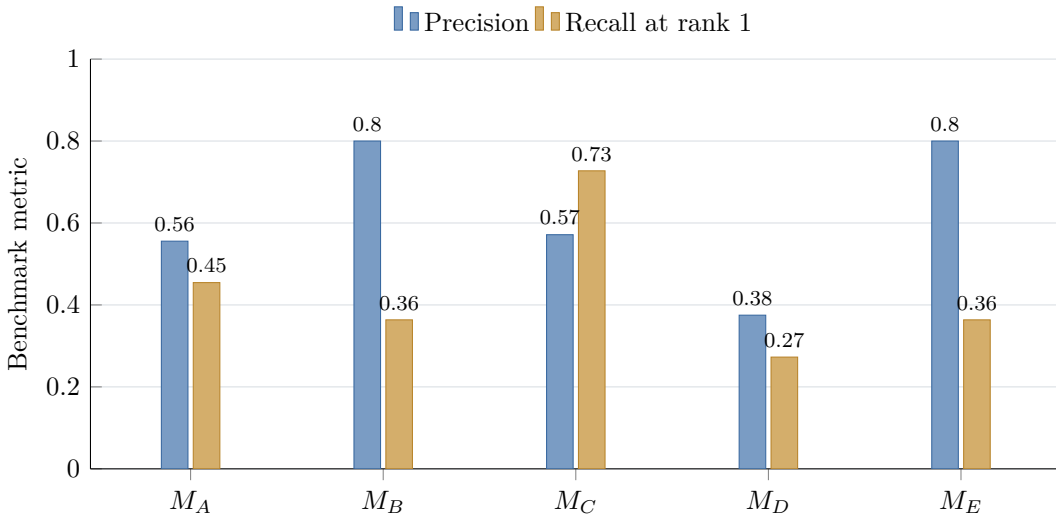


Figure 2: Precision and rank-one recall by implemented method on all evaluable benchmark anchors. The equal  $M_B$  and  $M_E$  bars mean the expiry rule does not alter the small benchmark, not that the two policies are identical on the full cohort.

### 5.4 Why each method works or fails

- **$M_A$  is rigid but not sufficiently precise.** Validated buyer identity and broad CPV continuity still occur across unrelated procurements from active buyers. It misses links without SIREN or CPV evidence and still links 8 of 63 reviewed negative anchors.
- **$M_B$  succeeds through repeated need-specific language.** Same-buyer blocking removes the hardest identity problem, while high local TF–IDF similarity captures repeated procurement subjects. Its 0.70 floor gives the best reviewed precision and zero negative-anchor FPR, but abstention limits recall to 0.3636.
- **$M_C$  recovers positives but over-links.** It reaches recall 0.7273, but accepts 10 of 63 reviewed negative anchors. Buyer evidence is common by design after blocking, broad CPV continuity is not unique to one contract, and score renormalisation can leave fewer observed dimensions. It remains useful only as a high-recall sensitivity arm.
- **$M_D$  converges numerically but is not calibrated.** The primary EM fit estimates  $\lambda = 0.05267$ , far above the real successor-pair rarity, while the free- $u$  robustness fit reaches the 0.5 guardrail. No posterior exceeds 0.90. The model learns dominant text and documentation

structures rather than the extremely rare successor class; conditional dependence between fields also violates its simplifying assumption. It achieves precision 0.3750 and is not adopted.

- $M_E$  **targets a full-cohort error pattern absent from the benchmark.** It preserves  $M_B$  metrics while changing 42 anchors in the complete application. This supports non-degradation, but manual review is required to establish whether those changes improve real precision.

## 6 Expiry-Aware Selection Method

### 6.1 Broad temporal retrieval remains unchanged

The candidate-generation interval is intentionally broad. A pair  $(a, c)$  is temporally eligible for the existing pool when

$$A_a + 90 \text{ days} \leq C_c \leq A_a + 2920 \text{ days.} \quad (13)$$

Equation 13 is a retrieval window, not an expiry window. It avoids missing early re-tenders, delayed replacements, or successors whose contract dates are unavailable. Expiry evidence is applied during selection rather than blocking.

### 6.2 Expected end follows a fixed evidence hierarchy

The best available expected end is resolved as

$$E_a = \begin{cases} F_a, & \text{if exactly one valid explicit end date exists,} \\ S_a + D_a, & \text{if one start date and reliable duration exist,} \\ A_a + D_a, & \text{if reliable duration exists,} \\ \text{missing,} & \text{otherwise.} \end{cases} \quad (14)$$

An explicit end is valid only when it is later than the award-notice date. Multiple distinct start or end dates are not averaged. A duration is used only when episode reconstruction reports `typed_consensus`. Missing duration is not replaced by a four-year assumption or any other default.



Figure 3: Expected-end evidence among 3,800 cohort contracts. Only 955 contracts (25.1%) receive a resolved expected end; unavailable expiry preserves the baseline text threshold.

### 6.3 Timing is measured relative to expected end

For every candidate with a resolved anchor expiry,

$$R_{ac} = C_c - E_a. \quad (15)$$

Negative  $R_{ac}$  means the candidate was published before expected expiry. The implemented classes are:

$$\text{class}(a, c) = \begin{cases} \text{expiry\_unknown}, & E_a \text{ is missing,} \\ \text{early\_more\_than\_1y}, & R_{ac} < -365, \\ \text{pre\_expiry\_within\_1y}, & -365 \leq R_{ac} < 0, \\ \text{post\_expiry\_within\_1y}, & 0 \leq R_{ac} \leq 365, \\ \text{post\_expiry\_more\_than\_1y}, & R_{ac} > 365. \end{cases} \quad (16)$$

The boundary is exact: a candidate published exactly 365 days before expected expiry uses the normal rule. Only candidates earlier than that boundary receive the stronger gate.

#### 6.4 Very early candidates require stronger functional evidence

Define the eligibility indicator  $I_{ac}$  as

$$I_{ac} = \begin{cases} \mathbf{1}[s_t(a, c) \geq 0.85 \wedge s_k(a, c) > 0], & R_{ac} < -365, \\ \mathbf{1}[s_t(a, c) \geq 0.70], & \text{otherwise.} \end{cases} \quad (17)$$

The second branch includes missing expiry. Missing CPV cannot pass the unusually early branch because missing evidence is not positive continuity. Among eligible candidates, ranking is deterministic: descending text similarity, descending existing linkage score, ascending candidate origin date, then ascending candidate episode identifier. One top candidate is selected per anchor, or the method abstains. This is a fixed policy, not a newly fitted model.

## 7 Results: Fewer Implausibly Early Events Without Benchmark Loss

### 7.1 The policy changes 42 baseline anchors

Table 4 compares the frozen baseline with the expiry-aware result. The new method accepts 40 fewer links overall. Of 547 baseline links, 505 are retained exactly. Forty are removed without replacement and two anchors receive a different selected candidate.

Table 4: Cohort-level comparison of frozen v1 and expiry-aware v2.

| Measure                  | V1 baseline  | Expiry-aware | Change       |
|--------------------------|--------------|--------------|--------------|
| Cohort contracts         | 3,800        | 3,800        | 0            |
| Accepted successor links | 547          | 507          | -40          |
| Censored contracts       | 3,253        | 3,293        | +40          |
| Event rate               | 14.39%       | 13.34%       | -1.05 pp     |
| Median time to successor | 32.07 months | 34.40 months | +2.33 months |

The direction is consistent with the intended correction: removing candidates that appear implausibly early lowers the estimated event rate and moves the median event later. These changes are descriptive effects of the selection policy, not evidence that the true renewal process changed.

The 42 changed anchors are fully represented in the review artifact. Twenty-three fail both the 0.85 text floor and CPV continuity, 11 fail only the text floor, 7 fail only CPV continuity, and one changes because deterministic tie-breaking selects another eligible candidate.

## 7.2 Most accepted links still have unknown expiry

Figure 4 reports the timing class of the 507 accepted links. Expiry remains unknown for 414 links (81.7%). The expiry-aware rule therefore improves a targeted subset of the linkage surface; it does not transform the full problem into expiry-based matching.

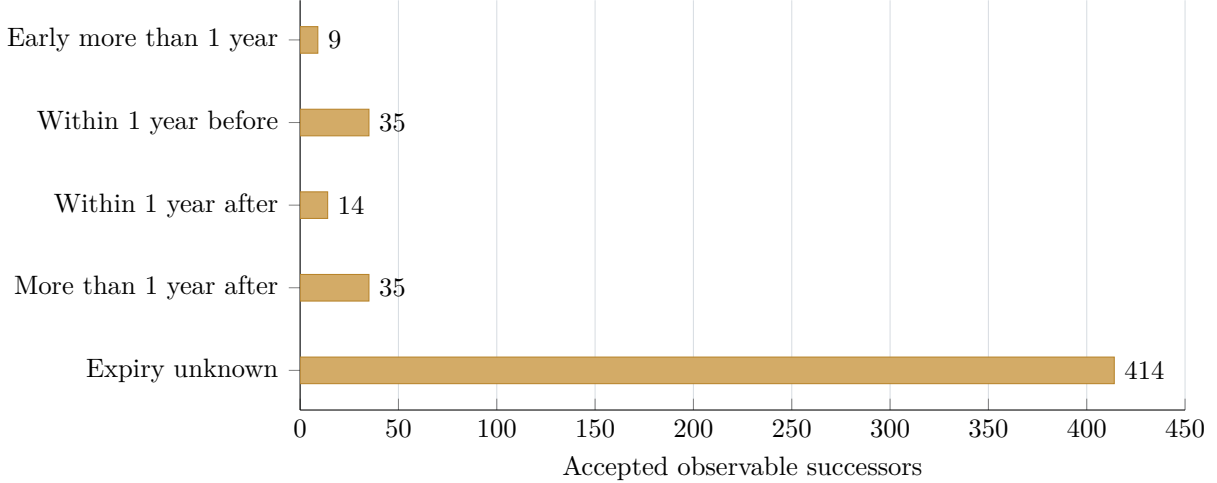


Figure 4: Timing classes of the 507 accepted expiry-aware links. Nine very early links remain because they satisfy both the 0.85 text threshold and positive CPV continuity.

## 7.3 The reviewed benchmark is unchanged

The benchmark contains 85 evaluable anchors: 22 reviewed positive anchors and 63 reviewed negatives. Table 5 shows that the expiry-aware policy makes no rank-one change on this benchmark. Both methods select 10 links: eight are correct and two choose the wrong successor on positive anchors. Neither method links a reviewed negative anchor.

Table 5: Benchmark comparison on all 85 evaluable anchors.

| Metric                             | V1 text ranking | Expiry-aware |
|------------------------------------|-----------------|--------------|
| Accepted benchmark links           | 10              | 10           |
| True positives                     | 8               | 8            |
| Wrong successor on positive anchor | 2               | 2            |
| False positive on negative anchor  | 0               | 0            |
| Precision at rank 1                | 0.8000          | 0.8000       |
| Recall at rank 1                   | 0.3636          | 0.3636       |
| Recall at rank 5                   | 0.4091          | 0.4091       |
| False-positive rate on negatives   | 0.0000          | 0.0000       |

The locked subset is also unchanged: precision 0.8750, recall 0.4118, and negative FPR 0.0000 across 69 anchors. These metrics demonstrate non-degradation on the available review data. They do not demonstrate higher population accuracy because the benchmark is small, was conditioned on an earlier ranker’s top-25 candidates, and was inspected during method development.

#### 7.4 Every digital segment retains adequate event volume

Figure 5 compares event counts by primary digital CPV segment. The largest absolute change occurs in CPV-72 IT services, from 185 to 167 events. Every segment remains above 30 events, preserving the minimum event volume used by the project’s stratified Kaplan–Meier analysis.

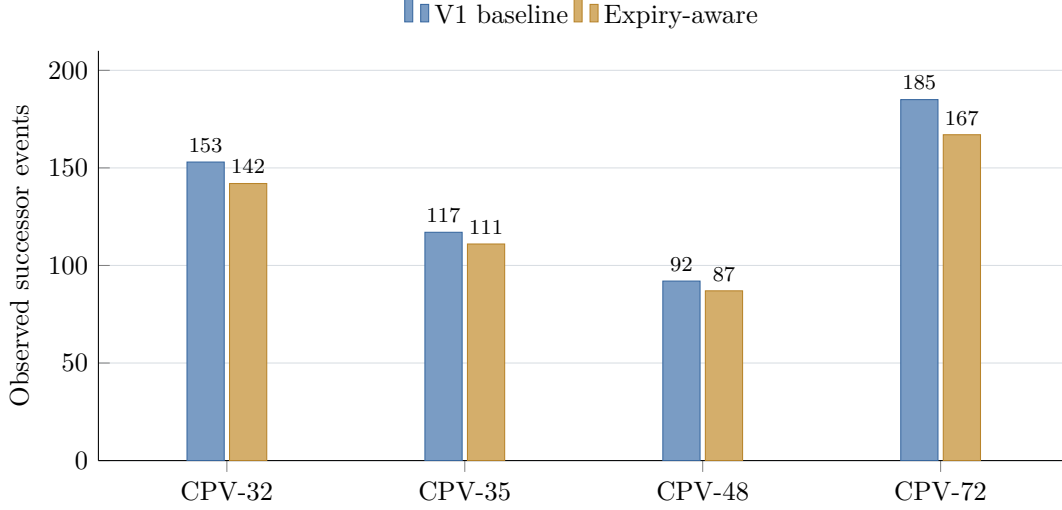


Figure 5: Observed successor events by digital segment. Counts fall in every segment, but all remain above the 30-event analysis minimum.

Regional event counts also fall consistently: Bretagne 166 to 152, Normandie 160 to 150, and Pays de la Loire 221 to 205. The change is not isolated to one region.

## 8 Survival Dataset and Validation

For each anchor  $a$ , the expiry-aware survival row is

$$(\delta_a, T_a) = \begin{cases} (1, C_{\hat{y}_a} - A_a), & \hat{y}_a \neq \emptyset, \\ (0, C_{\text{cutoff}} - A_a), & \hat{y}_a = \emptyset, \end{cases} \quad (18)$$

where  $C_{\text{cutoff}}$  is 31 December 2025. A censored row means no accepted observable successor before the cutoff; it does not prove that renewal never occurred.

Table 6: Structural validation of the expiry-aware survival dataset.

| Check                                     | Result |
|-------------------------------------------|--------|
| Rows                                      | 3,800  |
| Unique anchor episodes                    | 3,800  |
| Observed events                           | 507    |
| Right-censored rows                       | 3,293  |
| Negative durations                        | 0      |
| Zero-duration rows                        | 3      |
| Duplicate anchor episodes                 | 0      |
| Successor equals anchor                   | 0      |
| All four segments with at least 30 events | Yes    |
| Automated validation passed               | Yes    |

The three zero-duration rows are contracts awarded on the observation cutoff with no accepted successor; they have no follow-up time and are not negative-duration errors.

Documentation-quality diagnostics remain relevant. Among linked anchors, 29.19% have a trusted SIREN versus 34.35% among unlinked anchors. Reliable duration is present for 18.34% of linked anchors versus 26.12% of unlinked anchors. Framework contracts are more common among linked anchors (24.06%) than unlinked anchors (19.13%). These are descriptive differences, not causal effects.

## 9 Promotion Gate, Limitations, and Robustness

### 9.1 The numerical gate passes

The fixed promotion checks are:

1. benchmark precision is at least 0.80;
2. benchmark false-positive rate on negatives is no higher than v1;
3. rank-one recall loses no more than one of 22 confirmed positive anchors.

All three checks pass. The recorded recommendation is `review_then_consider_promotion`; manual review remains mandatory.

### 9.2 Evidence limitations prevent an automatic claim of superiority

- **Expiry sparsity.** Expected end is unavailable for 74.9% of cohort contracts and for 414 of 507 accepted links. Those links still use the baseline text threshold.
- **Expected end is not always legal expiry.** Only 86 cohort contracts use one valid explicit end date. Most resolved dates are calculated from duration and either start or award date.
- **The benchmark is small.** Only 22 evaluable positive anchors exist, so one outcome changes recall by 4.55 percentage points.
- **The benchmark is ranker-conditioned.** Reviewers saw candidates from a previous top-25 export, so buried successors were unavailable for labelling.
- **The locked split is not pristine.** It was inspected during earlier development and is evaluation evidence rather than an untouched holdout.
- **Policy thresholds are judgement rules.** The 365-day boundary and 0.85 early text threshold target an observed failure mode; they were not estimated from sufficient labelled

data.

- **Successor is not renewal.** A selected pair can represent re-tendering, replacement, continuation, or a recurrent need. BOAMP has no legal renewal ID.

The evidence supports the claim that the new method is more conservative and better aligned with contract timing. It does not establish true population precision or prove that every selected link is a renewal.

## 10 Recommended Decision and Next Steps

The expiry-aware method should remain the preferred promotion candidate and v1 should remain the reproducible baseline until review is complete. The next work is narrow:

1. Review all 42 changed anchors in `data/processed/boamp_v2/expiry_link_review.csv`.
2. Assign one label: correct removal, incorrect removal, correct replacement, incorrect replacement, or insufficient evidence.
3. Promote only if review supports the early-evidence gate. If genuine early successors are frequently removed, revise the boundary or gate and rerun the same parallel evaluation.
4. If promoted, rerun Kaplan–Meier, Cox, parametric, and watchlist outputs using the expiry-aware survival dataset while retaining v1 as sensitivity evidence.

No additional machine-learning model is recommended before this review. The highest value next labels are the 42 cases where the policy changes the event definition.

## 11 Further Questions

- How many of the 40 removed links are genuinely unrelated concurrent procurements?
- Do explicit end dates and calculated expected ends have different error rates?
- Should candidates more than one year after expected expiry also receive an additional functional-continuity gate?
- Does the substantive survival conclusion remain stable after a full rerun of Kaplan–Meier and regression models on the 507-event dataset?

## A Reproducibility and Materialised Artifacts

Table 7: Implementation and evidence inventory.

| Artifact                                                           | Role                                                                                           |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| boamp_pipeline/expiry_linkage.py                                   | Date resolution, timing classes, eligibility rule, and deterministic ranking.                  |
| scripts/evaluate_expiry_aware_linkage.py                           | Candidate enrichment, benchmark comparison, promotion gate, accepted links, and review export. |
| scripts/build_expiry_aware_survival_dataset.py                     | Parallel right-censored survival dataset and validation summary.                               |
| tests/test_expiry_linkage.py                                       | Unit tests for date precedence, boundaries, missingness, gating, and deterministic selection.  |
| data/processed/boamp_v2/expiry_aware_linkage_summary.json          | Linkage policy, benchmark, timing, cohort comparison, and validation metrics.                  |
| data/processed/boamp_v2/expiry_link_review.csv                     | Complete 42-anchor manual review queue.                                                        |
| data/processed/boamp_v2/survival_dataset_expiry_aware.parquet      | Parallel 3,800-row survival dataset with 507 events.                                           |
| data/processed/boamp_v2/survival_dataset_expiry_aware_summary.json | Survival validation, segment results, regional results, and linkage-bias diagnostics.          |

Reproduction commands:

```
PYTHONPATH=. python3 scripts/evaluate_expiry_aware_linkage.py --force
PYTHONPATH=. python3 scripts/build_expiry_aware_survival_dataset.py --force
PYTHONPATH=. pytest -q
```

At report generation, the test suite reports 22 passing tests. The linkage and survival summary artifacts both report `validation_passed: true`. The v1 candidate pool, linkage outputs, and survival outputs are not overwritten.

## B Output Schema Additions

The accepted-link artifact preserves existing candidate fields and adds:

- `anchor_expected_end_date`;
- `expected_end_source`;
- `explicit_start_date` and `explicit_end_date`;
- `explicit_start_date_count` and `explicit_end_date_count`;
- `days_from_expected_end`;
- `timing_class`;
- `selection_method`;
- `expiry_linkage_version`.

The parallel survival artifact carries timing and selection fields for linked events while retaining the existing cohort grain and censoring definition.